Experiment ‘p_6_1_d_process_array_first_and_last_array’ Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_1_d_process_array_first_and_last_array
Natural language statement: Write a method that takes an array of ten integers and then returns the first and last elements.
Method signature: p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (first_and_last: array<int>)

Ensures

- `first_and_last[..] == [inputs[0], inputs[inputs.Length - 1]]`

Requires

- `inputs.Length == 10`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes an array of ten integers and then returns the first and last
    ↪ elements.

The signature should be:

p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (first_and_last:
    ↪ array<int>)

The method should respect the following contract:

requires inputs.Length == 10, ensures first_and_last[..] == [inputs[0], inputs[inputs.
    ↪ Length - 1]]

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn0AVNSETutUUHx8KMOEXFvgX1Qrs
Model: gpt-5.5-2026-04-23
Created at: 1780571091
Finish reason: stop
Usage: 322 tokens in, and 290 tokens out

```
{
  first_and_last := new int[2];
  first_and_last[0] := inputs[0];
  first_and_last[1] := inputs[inputs.Length - 1];
}
```

Final Program

```
method p_6_1_d_process_array_first_and_last_array(inputs: array<int>) returns (
    ↪ first_and_last: array<int>)
      requires inputs.Length == 10
      ensures first_and_last[..] == [inputs[0], inputs[inputs.Length - 1]]
{
  first_and_last := new int[2];
  first_and_last[0] := inputs[0];
  first_and_last[1] := inputs[inputs.Length - 1];
```

```
}
```

Total Token Usage

Input tokens: 322
Output tokens: 290
Reasoning tokens: 226
Sum of 'total tokens': 612

Experiment Timings

Overall Experiment started at 1780571091575, ended at 1780571120846, lasting 29271ms (29.27 seconds)
Iteration #1 started at 1780571091606, ended at 1780571120846, lasting 29240ms (29.24 seconds)